

1. Are these relations well-structured? Explain your reasoning.

The relations seem well structured this is because most tables have a primary key and the data can all be traced back to the needed data. The tables also have a limited amount of data which means that the tables don't contain too many values and reference the other tables making look ups easy

2. Identify the primary Key(s) for each table

Buildings: building_ID

Apartments: apartment_ID

Amenities: amenity_ID

Building Amenities: amenity_ID, building_ID

3. Identify any foreign keys and explain what they reference.

Buildings building_ID → Apartments

Apartments: building_ID → Buildings

Amenities: amenity_ID → Building Amenities

Building amenities: amenity_ID → Amenities

4. Based on the data model, how many apartments can a building have?

One to Many

5. How many buildings is each apartment associated with?

Each apartment is associated with only 1 building

6. Based on the structure of the tables, how many amenities can each building be associated with?

Each building can be associated with up to 5 amenities

7. Can the same amenity be associated with more than one building? Explain how you know.

Yes, the same amenity can be associated with more than one building, I know this because in the building amenities table separate buildings are associated with the same amenities

8. Based on the information you see, which building has apartments with 3 bedrooms?

Maple heights are the building that has an apartment with 3 bedrooms

9. Based on the information you see, is the Riverfront Tower pet friendly?

Yes, it is pet friendly

10. If you wanted to add information about apartment availability (possibly just an indicator on whether apartments are available or even the date that they become available), which table and how would you add this information?

I would add this to the apartments table because it holds the information about the state of the apartment and the vacancy would just be another attribute in that table

11. Identify possible Entity types and relationships between them and use oracle sql developer to draw a preliminary ER diagram. At this point all you need to draw is a box-and-lines diagram.

